GGG-D-647a
September 21, 1965

SUPERSEDING
Int. Fed. Spec. GGG-D-00647 (GSA-FSS)
April 24, 1964

FEDERAL SPECIFICATION

DRILL BLANKS AND DRILL BLANK SETS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers drill blanks and drill blank sets. The drill blanks are suitable for use as plug gages and can be machined into cutting tools (see 6.1).

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Type.* The drill blanks shall be of the type and size as specified in 3.3. Type I is letter sizes, type II is fractional sizes, and type III is wire gage sizes (see 6.2). In addition to type designation, sets shall be indicated as follows:

Set No. 1—Letter size. A through Z.

Set No. 2—Fractional size, 1/16 inch through 1/2 inch by 64ths.

Set No. 3—Wire gage sizes, 1 through 60.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

GGG-C-116—Case, Twist Drill Set; And Stand, Twist Drill Set—Empty.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 151—Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools, Packaging Of.

Military Standards:

MIL-STD-10—Surface Roughness, Waviness and Lay.

FSC 5133

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MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Illustration.** The illustration herein is descriptive and not restrictive and is not intended to preclude the purchase of drill blanks, which are otherwise in accordance with this specification.

3.2 **Material.** Drill blanks shall be made of H. S. steel conforming to AISI-SAE standard for M-1, M-2, or M-10 steel.

3.2.1 *Hardness.* Drill blanks shall have a Rockwell "C" hardness of 62 to 66.

3.3 **Dimensions and tolerances.** The dimensions and tolerances shall conform to table I, II, or III as required.

TABLE I. *Type I drill blanks. (Plus series)*

Letter size	Diameter inches +0.0002 −0.0000	Length overall inches ± 1/8
A	0.2340	3-7/8
B	.2380	4
C	.2420	4
D	.2460	4
E	.2500	4
F	.2570	4-1/8
G	.2610	4-1/8
H	.2660	4-1/8
I	.2720	4-1/8
J	.2770	4-1/8
K	.2810	4-1/4
L	.2900	4-1/4
M	.2950	4-3/8
N	.3020	4-3/8
O	.3160	4-1/2
P	.3230	4-5/8
Q	.3320	4-3/4
R	.3390	4-3/4
S	.3480	4-7/8
T	.3580	4-7/8
U	.3680	5
V	.3770	5
W	.3860	5-1/8
X	.3970	5-1/8
Y	.4040	5-1/4
Z	.4130	5-1/4

Note: Blanks shall be concentric within 0.0002 inch. Blanks shall be straight (parallel) within 0.0005 inch.

TABLE II. *Type II drill blanks. (Plus series)*

Fractional size	Diameter inches +0.0002 −0.0000	Length overall inches ± 1/8
1/16	0.0625	1-7/8
5/64	.0781	2
3/32	.0937	2-1/4
7/64	.1094	2-5/8
1/8	.1250	2-3/4
9/64	.1406	2-7/8
5/32	.1562	3-1/8
11/64	.1718	3-1/4
3/16	.1875	3-1/2
13/64	.2031	3-5/8
7/32	.2187	3-3/4
15/64	.2344	3-7/8
1/4	.2500	4
17/64	.2656	4-1/8
9/32	.2812	4-1/4
19/64	.2968	4-3/8
5/16	.3125	4-1/2
21/64	.3281	4-5/8
11/32	.3437	4-3/4
23/64	.3593	4-7/8
3/8	.3750	5
25/64	.3906	5-1/8
13/32	.4062	5-1/4
27/64	.4218	5-3/8
7/16	.4375	5-1/2
29/64	.4531	5-5/8
15/32	.4687	5-3/4
31/64	.4843	5-7/8
1/2	.5000	6

Note: Blanks shall be concentric within 0.0002 inch. Blanks shall be straight (parallel) within 0.0005 inch.

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TABLE III. *Type III drill blanks.*
(Plus series)

Wire size	Diameter inches +0.0002 −0.0000	Length overall inches ± 1/8
1	0.2280	3-7/8
2	.2210	3-7/8
3	.2130	3-3/4
4	.2090	3-3/4
5	.2055	3-3/4
6	.2040	3-3/4
7	.2010	3-5/8
8	.1990	3-5/8
9	.1960	3-5/8
10	.1935	3-5/8
11	.1910	3-1/2
12	.1890	3-1/2
13	.1850	3-1/2
14	.1820	3-3/8
15	.1800	3-3/8
16	.1770	3-3/8
17	.1730	3-3/8
18	.1695	3-1/4
19	.1660	3-1/4
20	.1610	3-1/4
21	.1590	3-1/4
22	.1570	3-1/8
23	.1540	3-1/8
24	.1520	3-1/8
25	.1495	3
26	.1470	3
27	.1440	3
28	.1405	2-7/8
29	.1360	2-7/8
30	.1285	2-3/4
31	.1200	2-3/4
32	.1160	2-3/4
33	.1130	2-5/8
34	.1110	2-5/8
35	.1100	2-5/8
36	.1065	2-1/2
37	.1040	2-1/2
38	.1015	2-1/2
39	.0995	2-3/8
40	.0980	2-3/8
41	.0960	2-3/8
42	.0935	2-1/4
43	.0890	2-1/4
44	.0860	2-1/8
45	.0820	2-1/8
46	.0810	2-1/8
47	.0785	2
48	.0760	2
49	.0730	2
50	.0700	2
51	.0670	2
52	.0635	1-7/8
53	.0595	1-7/8
54	.0550	1-7/8
55	.0520	1-7/8
56	.0465	1-3/4
57	.0430	1-3/4
58	.0420	1-5/8
59	.0410	1-5/8
60	.0400	1-5/8

Note: Blanks shall be concentric within 0.0002 inch. Blanks shall be straight (parallel) within 0.0005 inch.

3.4 Design. Drill blanks shall have no back taper and be similar to figure 1. Cone shaped ends are optional with the supplier. If cone shaped ends are furnished, they shall have a 118° ± 1° included angle. Ends that are not cone shaped shall be at right angle to the centerline of the blank within ± 3°.

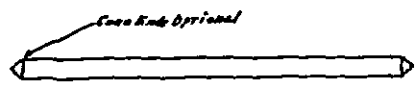


FIGURE 1. Types, I, II, and III drill blanks.

3.5 Finish. Drill blanks shall have a maximum surface roughness of 32 microinches r.h.r. or better conforming to requirements in MIL-STD-10. Surfaces shall be free of metal defects such as, inclusions, porosity, segregation, flakes, cracks, oxidation, or foreign matter.

3.5.1 *Coating.* The finished drill blank shall be coated with a thin coating of rust preventative oil.

3.6 Drill blanks. When drill blanks are to be furnished separately, the size of each blank shall be specified. Sizes shall be taken from applicable tables (see 6.2).

3.7 ***Drill blank sets.*** When sets are required, the drill blanks shall be furnished as specified (see 6.2). Drill blank sets shall be furnished in a case conforming to requirements of GGG-C-116; type I, class 1,

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size D for letter size drill blanks A through Z; type I, class 1, size E for fractional size drill blank 1/16 inch through 1/2 inch by 64th inch; type I, class 1, size A for numbered drill blanks 1 through 60.

3.8 **Identification marking.** Marking of the case shall conform to requirements of 3.6 of GGG-C-116. Each drill blank 0.125 and larger shall be permanently marked with the suppliers trademark or symbol, and size designation.

3.9 **Workmanship.** Drill blanks shall be free of defects that would be detrimental to usability or serviceability, and the quality of the workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 **Sampling procedures.** Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table IV.

4.2.1 *Drill blank sets.* Each component of the set shall be formed into a sublot. The cases for sets shall be inspected in accordance with GGG-C-116.

4.3 **Examination.**

4.3.1 *Visual examination.* Each sample unit shall be examined for any nonconformance in design, material, finish, coating, workmanship, and marking.

4.3.2 *Dimensional examination.* Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 **Testing.** Each sample unit shall be tested in accordance with 4.4.1.

4.4.1 *Testing.* Each sample unit shall be tested for hardness in accordance with method 243.1 of Fed. Test Method Std. No. 151 to determine compliance with 3.2.1.

4.5 **Inspection of preparation for delivery requirements.** An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table V. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The lot size shall be the

TABLE IV. *Sampling data*

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 ea.	II	4.0	Defects per hundred units	4.3.1
Dimensional examination ..	1 ea.	S-4	1.0	Defects per hundred units	4.3.2
Testing	1 ea.	S-4	1.0	Percent defective	4.4

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number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE V. *Classification of preparation for delivery defects*

Examine	Defects
Markings, (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Preservation, packaging, and packing. Unless otherwise specified (see 6.2), the drill blanks and drill blank sets shall be preserved, packaged, and packed in accordance with MIL-H-15424. The level of preservation and packaging shall be A or C and the level of packing A, B, or C as specified (see 6.2).

5.1.1 As part of the requirements of section 5, the following is considered as part of table I of MIL-H-15424:

Col. 1	Col. 2	Col. 3	Col. 4	Col. 5	Col. 6	Col. 7
Blanks: Drill	P-2, P-7, P-8, P-9	I	12	12	16	2304
Set: Drill blanks	P-8, P-9, P-10	IA	1		36	36

5.2 Marking.

5.2.1 *Civil agencies.* In addition to markings required by the contract or order, the unit packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to markings required by the contract or order, the unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Drill blanks can be adapted for use as plug gages, and machined into tool bits, routers, countersinks, counterbores, drills, reamers, end mills, shaper tools, and the like.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Drill blanks in sets. Specify type and set numbers (see 1.2.1).
(*c*) Drill blanks not in sets. Specify type and size of each blank required (see 1.2.1).
(*d*) Level of preservation, packaging, and packing required (see 5.1).
(*e*) Special marking, if required, (see 5.2).

6.3 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:
Tool bit blanks, steel.
Carload minimum weight 30,000 pounds.

Motor:
Tool bit blanks, steel.
Carload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification

DOD coordination has been waived.
Preparing Activity:
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